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Real-time misfire detection

Abstract

Systems and methods for wireless power transfer systems are described. A controller can be coupled to a power rectifier configured to rectify alternating current power into direct current power. The power rectifier can include a first high side transistor, a second high side transistor, a first low side transistor, and a second low side transistor. The controller can be configured to operate the power rectifier in full bridge rectifier mode. The controller can be further configured to detect a misfire event between the wireless power receiver and a wireless power transmitter. The controller can be further configured to, in response to detection of the misfire event, switch an operation mode of the power rectifier from the full bridge rectifier mode to a diode mode.

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Background/Summary

BACKGROUND

(1) The present disclosure relates in general to apparatuses and methods for operating wireless power receivers under different modes to address misfire events in wireless power transfer applications.

(2) Wireless power systems can include a transmitter having a transmission coil and a receiver having a receiver coil. In an aspect, the transmitter may be connected to a structure including a wireless charging region. In response to a device including the receiver being placed on the charging region, or in proximity to the charging region, the transmission coil and the receiver coil can be inductively coupled with one another to form a transformer that can facilitate inductive transfer of alternating current (AC) power. The transfer of AC power, from the transmitter to the receiver, can facilitate charging of a battery of the device including the receiver.

SUMMARY

(3) In one embodiment, a semiconductor device for wireless power receiver is generally described. The semiconductor device can include a power rectifier and a controller. The power rectifier can be configured to rectify alternating current (AC) power into direct current (DC) power. The power rectifier can include a first high side transistor, a second high side transistor, a first low side transistor, and a second low side transistor. The controller can be configured to operate the power rectifier in full bridge rectifier mode. The controller can be further configured to detect a misfire

event between the wireless power receiver and a wireless power transmitter. The controller can be further configured to, in response to the detection of the misfire event, switch an operation mode of the power rectifier from the full bridge rectifier mode to a diode mode.

(4) In one embodiment, an apparatus for operating a wireless power receiver is generally described. The apparatus can include a plurality of comparators and a controller. The plurality of comparators can be configured to measure voltage levels across a first high side transistor, a second high side transistor, a first low side transistor, and a second low side transistor of a power rectifier. The plurality of comparators can be further configured to output a plurality of signals. The controller can be configured to operate the power rectifier in full bridge rectifier mode. The controller can be further configured to, based on the plurality of signals outputted by the plurality of comparators, detect a misfire event between the wireless power receiver and a wireless power transmitter. The controller can be further configured to, in response to the detection of the misfire event, switch an operation mode of the power rectifier from the full bridge rectifier mode to a diode mode.

(5) In one embodiment, a method for operating a wireless power receiver is generally described. The method can include operating, by a controller, a power rectifier of a wireless power receiver in full bridge rectifier mode. The method can further include receiving, by the controller, a plurality of signals from a plurality of comparators. The plurality of signals can be based on measurements of voltage levels across a first high side transistor, a second high side transistor, a first low side transistor, and a second low side transistor of a power rectifier. The method can further include, based on the plurality of signals, detecting, by the controller, a misfire event between the wireless power receiver and a wireless power transmitter. The method can further include, in response to the detection of the misfire event, switching, by the controller, an operation mode of the power rectifier from the full bridge rectifier mode to a diode mode.

(6) The foregoing summary is illustrative only and is not intended to be in any way limiting. In addition to the illustrative aspects, embodiments, and features described above, further aspects, embodiments, and features will become apparent by reference to the drawings and the following detailed description. In the drawings, like reference numbers indicate identical or functionally similar elements.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a diagram showing an example system that can implement real-time misfire detection in one embodiment.
- (2) FIG. 2 is a diagram showing a circuit that can implement real-time misfire detection in one embodiment.
- (3) FIG. 3 is a diagram showing a circuit that can implement real-time misfire detection in one embodiment.
- (4) FIG. 4 is a flow diagram illustrating a process of implementing real-time misfire detection in one embodiment.
- (5) FIG. 5 is a diagram illustrating waveforms of an operation of a wireless power receiver implementing real-time misfire detection in one embodiment.
- (6) FIG. 6 is a flow diagram illustrating another process of implementing real-time misfire detection in one embodiment.

DETAILED DESCRIPTION

(7) In the following description, numerous specific details are set forth, such as particular structures, components, materials, dimensions, processing steps and techniques, in order to provide an understanding of the various embodiments of the present application. However, it will be appreciated by one of ordinary skill in the art that the various embodiments of the present

application may be practiced without these specific details. In other instances, well-known structures or processing steps have not been described in detail in order to avoid obscuring the present application.

(8) FIG. 1 is a diagram showing an example system that can implement real-time misfire detection in one embodiment. System **100** can include power devices, such as a transmitter **110** and a receiver **120**, that are configured to wirelessly transfer power and data therebetween via inductive coupling. While described herein as transmitter **110** and receiver **120**, each of transmitter **110** and receiver **120** may be configured to both transmit and receive power or data therebetween via inductive coupling. Transmitter **110** can be referred to as a wireless power transmitter and receiver **120** can be referred to as a wireless power receiver.

(9) Transmitter **110** is configured to receive power from one or more power supplies and to transmit AC power **130** to receiver **120** wirelessly. For example, transmitter **110** may be configured for connection to a power supply **116** such as, e.g., an adapter or a DC power supply. Transmitter **110** can be a semiconductor device including a controller **112** and a power driver **114**.

(10) Controller **112** can be configured to control and operate power driver **114**. Controller **112** can include, for example, a processor, central processing unit (CPU), field-programmable gate array (FPGA) or any other circuitry that is configured to control and operate power driver **114**. While described as a CPU in illustrative embodiments, controller **112** is not limited to a CPU in these embodiments and may comprise any other circuitry that is configured to control and operate power driver **114**. In an example embodiment, controller **112** can be configured to control power driver **114** to drive a coil TX of the power driver **114** to produce a magnetic field. Power driver **114** can be configured to drive coil TX at a range of frequencies and configurations defined by wireless power standards, such as, e.g., the Wireless Power Consortium (Qi) standard, the Power Matters Alliance (PMA) standard, the Alliance for Wireless Power (A for WP, or Rezence) standard or any other wireless power standards.

(11) Receiver **120** can be configured to receive AC power **130** transmitted from transmitter **110** and to supply the power to one or more loads **126** or other components of a destination device **140**. Load **126** may comprise, for example, a battery charger that is configured to charge a battery of the destination device **140**, a DC-DC converter that is configured to supply power to a processor, a display, or other electronic components of the destination device **140**, or any other load of the destination device **140**. Destination device **140** may comprise, for example, a computing device, mobile device, mobile telephone, smart device, tablet, wearable device or any other electronic device that is configured to receive power wirelessly. In an illustrative embodiment, destination device **140** can include receiver **120**. In other embodiments, receiver **120** may be separate from destination device **140** and connected to destination device **140** via a wire or other component that is configured to provide power to destination device **140**.

(12) Receiver **120** can be a semiconductor device including a controller **122**, a resonant circuit **123**, and a power rectifier **124**. Controller **122** can be an integrated circuit including, for example, a digital controller such as a microcontroller, a processor, CPU, FPGA or any other circuitry that may be configured to control and operate power rectifier **124**. Resonant circuit **123** can include a coil RX and one or more capacitors, inductors, resistors, that can form circuitry for outputting communication packets **136** and conveying AC power **130**, received from transmitter **110**, to power rectifier **124**. Power rectifier **124** can include rectifier circuits such as half-bridge rectifiers, full bridge rectifiers, or other types of rectifier circuits that can be configured to rectify power received via resonant coil RX of resonant circuit **123** into a power type as needed for load **126**. Power rectifier **124** is configured to rectify AC power **130** into DC power **132** which may then be supplied to load **126**. Controller **122** can be configured to execute application specific programs and/or firmware to control and operate various components, such as resonant circuit **123** and power rectifier **124**, of receiver **120**.

(13) As an example, when receiver **120** is placed in proximity to transmitter **110**, the magnetic field

produced by coil TX of power driver **114** induces a current in coil RX of resonant circuit **123**. The induced current causes AC power **130** to be inductively transmitted from power driver **114** to power rectifier **124**, via resonant circuit **123**. Power rectifier **124** receives AC power **130** and converts AC power **130** into DC power **132**. DC power **132** is then provided by power rectifier **124** to load **126**.

(14) Transmitter **110** and receiver **120** are also configured to exchange information or data, e.g., messages, via the inductive coupling of power driver **114** and resonant circuit **123**. For example, before transmitter **110** begins transferring power to receiver **120**, a power contract may be agreed upon and created between receiver **120** and transmitter **110**. For example, receiver **120** may send communication packets **136** or other data to transmitter **110** that indicate power transfer information such as, e.g., an amount of power to be transferred to receiver **120**, commands to increase, decrease, or maintain a power level of AC power **130**, commands to stop a power transfer, or other power transfer information. In another example, in response to receiver **120** being brought in proximity to transmitter **110**, e.g., close enough such that a transformer may be formed by coil TX and coil RX to allow power transfer, receiver **120** may be configured to initiate communication by sending a signal to transmitter **110** that requests a power transfer. In such a case, transmitter **110** may respond to the request by receiver **120** by establishing the power contract or beginning power transfer to receiver **120**, e.g., if the power contract is already in place. Transmitter **110** and receiver **120** may transmit and receive communication packets **136**, data or other information via the inductive coupling of coil TX and coil RX. In some embodiments, communication between transmitter **110** and receiver **120** can occur before power transfer stage using various protocols such as near field communication (NFC), Bluetooth, etc.

(15) Details of power rectifier **124** is shown in FIG. **1**. In an example, in response to receiving AC power **130**, power rectifier **124** can rectify AC power **130** into DC voltage in order to supply DC power **132** to load **126**. AC power **130** can be received through the AC1 and AC2 nodes of power rectifier **124**. Four transistors HS1, HS2, LS1, LS2, which can be metal-oxide-semiconductor field-effect transistors (MOSFETs), can form power rectifier **124** that enable supplying the rectifier voltage (VRECT) to rectify the AC power **130**. Controller **122** can alternately switch the pair of transistors HS1, LS1 and the pair of transistors HS2, LS2 to operate power rectifier **124** in an active mode (e.g., full bridge rectifier mode). Under the active mode, when transistors HS1, LS1 are switched on, transistors HS2, LS2 are switched off, and vice versa. Further, under the active mode, current can flow from ground to VRECT in order to supply VRECT. Controller **122** can also control power rectifier **124** to operate under a passive mode. The passive mode of power rectifier **124** can be a diode mode, where transistors HS1, HS2, LS1, LS2 are switched off and the intrinsic diodes of transistors HS1, HS2, LS1, LS2 can allow current to flow from ground to VRECT through intrinsic diodes of transistors HS1, HS2, LS1, LS2.

(16) An active mode of power rectifier **124** can start at an initial state where all transistors HS1, HS2, LS1, LS2 are switched off. From the initial state, controller **122** can switch on a pair of transistors, such as transistors HS1, LS1, to form a current path from ground to VRECT via transistors HS1, LS1 (herein “HS1-LS1 path”). When transistors HS1, LS1 are switched on, transistors HS2, LS2 are switched off. To switch from the HS1-LS1 path to another current path from ground to VRECT via transistors HS2, LS2 (herein “HS2-LS2 path”), controller **122** can wait for a minimum-ON time (“MIN-ON time”) before performing the switch. The MIN-ON time can be a minimum required time for a pair of transistors to stay switched on under active mode. In response to expiration of the MIN-ON time, controller **122** can switch off transistors HS1, LS1 to switch off all transistors HS1, HS2, LS1, LS2, then subsequently switch on the pair of transistors HS2, LS2 to form the HS2-LS2 path.

(17) To have a proper rectification, power rectifier **124** has to be precisely synchronized with transmitter **110**. Under particular conditions, the synchronization can be lost and this causes an overshoot in the coil current that is discharging VRECT—which can be referred to as a misfire

event. A misfire event can include one or more cycles where current goes from VRECT to ground, instead of from ground to VRECT. When current goes from VRECT to ground, VRECT is being discharged. Misfire events can occur in response to heavy external perturbances, such as heavy load step changes or coupling factor changes (e.g. when destination device **140** is physically moved). Such external perturbation can cause an incorrect pair of high side and low side transistors to be switched on (e.g., HS2, LS2 can be incorrectly switched on during a cycle where HS1, LS1 is supposed to be switched on).

(18) In an aspect, misfire events can be self-sustaining for a variable number of cycles, and current can increase during these cycles, causing increased current to go in an undesirable direction (e.g., from VRECT to ground). For example, a misfire event can include more than one misfire cycle where current flows from VRECT to ground. In one embodiment, controller **122** can be configured to detect a first misfire cycle (e.g., an initial misfire cycle) of a misfire event (e.g., synchronization lost) when power rectifier **124** is operating under active mode. In response to the detection of the first misfire cycle, controller **122** can switch power rectifier **124** into passive or diode mode operation to restore synchronization of power rectifier with transmitter **110**. Further, controller **122** can be configured to monitor outputs from a circuit **150** that can be integrated between power rectifier **124** and controller **122**. Circuit **150** can include a plurality of comparators configured to detect current and/or voltage events at power rectifier **124**. For example, circuit **150** can detect voltages across transistors LS1, LS2 and, in response to specific values of the detected current and/or voltages, output signals that can be read by controller **122**. Controller **122** can read the signals outputted by circuit **150** to detect misfire events and/or to determine whether to switch power rectifier **124** to diode mode.

(19) FIG. 2 is a diagram showing a circuit that can implement real-time misfire detection in one embodiment. In an example shown in FIG. 2, circuit **150** can include a plurality of comparators C1, C2, C3, C4. In one embodiment, comparators C1, C2, C3, C4 in circuit **150** can be Schmitt triggers. In one embodiment, controller **122** can detect misfire events based on signals LS1_ON, LS2_ON, LS1_OFF, LS2_OFF.

(20) Comparator C1 can receive a voltage level measured across transistor LS2 as inputs, and output a signal LS2_ON. In one embodiment, signal LS2_ON can have a voltage level representing a binary value (e.g., logic HIGH or LOW, or binary one or zero). Controller **122** can read the voltage level of signal LS2_ON. The voltage level of signal LS2_ON can indicate a current flow direction, such as from VRECT to ground or from ground to VRECT, of current flowing through transistor LS2.

(21) Comparator C3 can receive a voltage level measured across transistor LS1 as inputs, and output a signal LS1_ON. In one embodiment, signal LS1_ON can have a voltage level representing a binary value (e.g., logic HIGH or LOW, or binary one or zero). Controller **122** can read the voltage level of signal LS1_ON. The voltage level of signal LS1_ON can indicate a current flow direction, such as from VRECT to ground or from ground to VRECT, of current flowing through transistor LS1.

(22) FIG. 3 is a diagram showing a circuit that can implement real-time misfire detection in one embodiment. A circuit **300** shown in FIG. 3 can include power rectifier **124** and comparators C1, C2, C3, C4. Under active or full bridge rectifier mode, controller **122** can alternately switch the pair of transistors HS1, LS1 and the pair of transistors HS2, LS2. For instance, transistors HS1, LS1 can be switched on while transistors HS2, LS2 are switched off, and transistors HS1, LS1 can be switched off while transistors HS2, LS2 are switched on. In response to transistors HS1, LS1 being switched on, the HS1-LS1 path can be formed from ground to VRECT and current can flow from AC2 to AC1. In response to transistors HS2, LS2 being switched on, the HS2-LS2 path can be formed from ground to VRECT and current can flow from AC1 to AC2.

(23) Controller **122** can be configured to generate a drive signal SNS1 that can be used for switching on transistors HS1, LS1. Further, controller **122** can be configured to generate a drive

signal SNS2 that can be used for switching on transistors HS2, LS2. Drive signals SNS1 and SNS2 can be nonoverlapping signals (e.g., will not have the same values or voltage levels). In one embodiment, under the active mode, controller **122** can output drive signal SNS1, as V_HS1 and V_LS1, to drive transistors HS1, LS1, via gate driver **202**, using SNS1. Controller **122** can output drive signal SNS2, as V_HS2 and V_LS2, to drive transistors HS2, LS2, via gate driver **202**, using SNS2.

(24) Under passive or diode mode, controller **122** can switch off transistors HS1, HS2, LS1, LS2. In response to transistors HS1, HS2, LS1, LS2 being switched off, current can flow from ground to VRECT through intrinsic diodes of transistors HS1, HS2, LS1, LS2, hence always charging VRECT. Under the diode mode, drive signal SNS1, SNS2 will not be outputted such that the gates of transistors HS1, HS2, LS1, LS2 will not be driven and transistors HS1, HS2, LS1, LS2 can be switched off.

(25) FIG. **4** is a flow diagram illustrating a process of implementing real-time misfire detection in one embodiment. The process **400** may include one or more operations, actions, or functions as illustrated by one or more of blocks **402**, **404**, **406**, **408** and/or **410**. Although illustrated as discrete blocks, various blocks can be divided into additional blocks, combined into fewer blocks, eliminated, or performed in parallel, and/or performed in different order, depending on the desired implementation. Descriptions corresponding to FIG. **4** can reference components of FIG. **1-3**.

(26) Process **400** can be performed by a wireless power receiver, such as receiver **120** of FIG. **1**, during a full bridge rectifier mode of a power rectifier (e.g., power rectifier **124** in FIG. **1**). Process **400** can be performed for identifying a potential misfire event, and determining whether the identified potential misfire event is real (e.g., has occurred) or not (e.g., has not occurred). Process **400** can be performed for an active current path (e.g., current path that is on) for every operation cycle of power rectifier **124**. Each operation cycle of power rectifier **124**, under the full bridge rectifier mode, can have one active current path while the other current path is off. For example, the HS1-LS1 path can be on while the HS2-LS2 path is off, and vice versa. Process **400** can begin at block **402**, where a minimum ON (MIN_ON) time of an active current path (e.g., HS1-LS1 path or HS2-LS2 path) can expire.

(27) In response to the expiration of the MIN-ON time, process **400** can proceed to block **404**. At block **404**, a controller (e.g., controller **122**) of the wireless power receiver can detect or determine whether a drive signal (e.g., SNS1 or SNS2) being used to switch on the active current path is logic HIGH or logic LOW. The detection at block **404** can be performed during a predetermined time window that can begin in response to the expiration of the MIN-ON time at each operating cycle of power rectifier **124**. A duration of the time window can be programmable and can be based on a desired implementation of system **100**.

(28) Within the time window at block **404**, if the drive signal that switches on the active current path is HIGH (**404**: NO), then process **400** can proceed to block **408**, where the controller can determine that there is no occurrence of misfire events. The drive signal that switches on the active current path being HIGH can indicate that the active current path has been switched on during an entirety of the MIN-ON time and synchronization was not lost during the MIN-ON time, hence no misfire events.

(29) Within the time window at block **404**, if the drive signal that switches on the active current path is LOW (**404**: YES), then the controller can determine that there is a potential misfire event. The drive signal that switches on the active current path being LOW can indicate that either 1) the active current path was scheduled to be switched off in response to the expiration of the MIN-ON time, or 2) the active current path was switched off during the MIN-ON time. The active current path being switched off during the MIN-ON time can indicate that a misfire event (e.g., synchronization lost) may have occurred before the expiration of MIN-ON. In response to the controller determining that there is a potential misfire event (**404**: YES), process **400** can proceed to block **406**.

(30) At block **406**, the controller can receive a voltage level of a signal being outputted by an ON-comparator corresponding to the active current path. The ON-comparator corresponding to the active current path can be the comparator that is monitoring voltage level across a low side transistor of the active current path. For example, comparator **C1** can monitor voltage level across transistor **LS2** for the **HS2-LS2** path and comparator **C3** can monitor voltage level across transistor **LS1** for the **HS1-LS1** path. The controller can use the received voltage level to determine whether the potential misfire event (identified at block **404**) is real (e.g., has occurred) or not (e.g., has not occurred).

(31) The voltage level received by the controller can indicate a current flow direction across the low side transistor of the active current path. If current is flowing in the correct direction (e.g., ground to **VRECT**), then the voltage of an AC node (e.g., **AC1** or **AC2**) corresponding to the active current path can go below ground, and the ON-comparator corresponding to the active current path can be toggled from LOW to HIGH. Hence, if the voltage level of the signal received by the controller at block **406** indicates logic HIGH (**406: YES**), then the controller can determine that there is no misfire event (e.g., block **408**).

(32) If current is flowing in an incorrect direction (e.g., **VRECT** to ground), then the voltage of an AC node (e.g., **AC1** or **AC2**) corresponding to the active current path can be above ground, and the ON-comparator corresponding to the active current path can remain LOW. Hence, if the voltage level of the signal received by the controller at block **406** indicates logic LOW (**406: NO**), then the controller can determine that the potential misfire event is real (e.g., block **410**).

(33) FIG. 5 is a diagram illustrating waveforms of an operation of a wireless power receiver implementing real-time misfire detection in one embodiment. Descriptions corresponding to FIG. 4 can reference components of FIG. 1-4. In FIG. 5, power rectifier **124** can be operating under full bridge rectifier (FBR) mode during a time period **502**. During time period **502**, **VRECT** current can be positive (e.g., greater than zero amperes), indicating that current is flow in the correct direction from ground to **VRECT**. At a time **510**, a voltage at node **AC1** can go below zero ground (e.g., below zero volts) and the **VRECT** current can become negative, indicating that current is flowing in an incorrect direction from **VRECT** to ground. Controller **122**, based on process **400** of FIG. 4, can detect a misfire event at a time period **504**. In one embodiment, time period **504** can be one operation cycle of power rectifier **124**, and controller **122** can detect the first misfire cycle of the misfire event. The detection of the first misfire cycle can be performed during a time window, which can be a predetermined amount of time, that begins in response to the expiration of the MIN-ON time.

(34) In response to detecting this first misfire cycle, controller **122** can switch power rectifier **124** to operate in diode mode for one or more cycles. In the example shown in FIG. 5, power rectifier **124** can operate under diode mode during time period **506**, where time period **506** can include two operation cycles. To operate power rectifier **124** in diode mode, controller **122** can switch off transistors **HS1**, **HS2**, **LS1**, **LS2** during time period **506**. The number of operation cycles to operate power rectifier **124** can be programmable, and can be dependent on various factors, such as a desired implementation of system **100**, a status of various operating parameters of receiver **120**, predetermined parameters such as MIN-ON time, and/or other types of factors relating to operations of receiver **120**. In one embodiment, controller **122** can operate power rectifier **124** until synchronization is restored (e.g., by monitoring for a desired **VRECT** current for a predetermined amount of time or cycles). In response to expiration of time period **506**, synchronization is restored (e.g., current is consistently above zero) and controller **122** can switch power rectifier **124** to operate under FBR mode in time period **508**.

(35) The apparatuses and methods described herein can configure a wireless power receiver to address misfire events by restoring synchronization in response to detection of a first misfire cycle, instead of allow the misfire event to prolong for more than one cycle. The detection of the first misfire cycle, and inserting one or more cycles of diode mode, can restore the synchronization

before current in the receiver coil raising to an undesirable level. Therefore, the reduced duration of misfire events can prevent current spike and overshoot into the receiver coil, and can reduce a risk of damaging the receiver coil. Further, the apparatuses and methods described herein may not require addition of relatively bulky analog hardware.

(36) FIG. 6 is a flow diagram illustrating another process of implementing real-time misfire detection in one embodiment. The process **600** may include one or more operations, actions, or functions as illustrated by one or more of blocks **602**, **604**, **606**, and/or **608**. Although illustrated as discrete blocks, various blocks can be divided into additional blocks, combined into fewer blocks, eliminated, or performed in parallel, and/or performed in different order, depending on the desired implementation. Descriptions of process **600** can refer to components of FIGS. 1-4.

(37) Process **600** can be performed by a controller (e.g., controller **122** described herein) of a wireless power receiver (e.g., receiver **120** described herein). Process **600** can begin at block **602**. At block **602**, a controller can operate a power rectifier of a wireless power receiver in full bridge rectifier mode. Process **600** can proceed from block **602** to block **604**. At block **604**, the controller can receive a plurality of signals from a plurality of comparators. The plurality of signals can be based on measurements of voltage levels across a first high side transistor, a second high side transistor, a first low side transistor, and a second low side transistor of a power rectifier.

(38) Process **600** can proceed from block **604** to block **606**. At block **606**, the controller can, based on the plurality of signals, detect a misfire event between the wireless power receiver and a wireless power transmitter. In one embodiment, the controller can detect the misfire event by detecting an initial misfire cycle of the misfire event.

(39) In one embodiment, under the full bridge rectifier mode of the power rectifier, the controller can, in response to an expiration of a minimum ON time, detect whether a drive signal being used for switching on a current path is HIGH or LOW. In response to the drive signal being HIGH, the controller can determine that the misfire event has not occurred. In response to the drive signal being LOW, the controller can identify a potential misfire event. In one embodiment, the controller can detect whether the drive signal is HIGH or LOW during a predetermined time window that begins in response to the expiration of the minimum ON time.

(40) In one embodiment, in response to identifying the potential misfire event, the controller can determine whether a voltage level measured across a low side transistor of the current path indicates HIGH or LOW. In response to the voltage level indicating HIGH, the controller can determine that the potential misfire event has not occurred. In response to the voltage level indicating LOW, the controller can determine that the potential misfire has occurred.

(41) Process **600** can proceed from block **606** to block **608**. At block **608**, the controller can, in response to the detection of the misfire event, switch an operation mode of the power rectifier from the full bridge rectifier mode to a diode mode. In one embodiment, the controller can switch the operation mode of the power rectifier from the full bridge rectifier mode to the diode mode by switching off the first high side transistor, the second high side transistor, the first low side transistor, and the second low side transistor.

(42) The flowchart and block diagrams in the Figures illustrate the architecture, functionality, and operation of possible implementations of systems, methods, and computer program products according to various embodiments of the present invention. In this regard, each block in the flowchart or block diagrams may represent a module, segment, or portion of instructions, which comprises one or more executable instructions for implementing the specified logical function(s). In some alternative implementations, the functions noted in the blocks may occur out of the order noted in the Figures. For example, two blocks shown in succession may, in fact, be executed substantially concurrently, or the blocks may sometimes be executed in the reverse order, depending upon the functionality involved. It will also be noted that each block of the block diagrams and/or flowchart illustration, and combinations of blocks in the block diagrams and/or flowchart illustration, can be implemented by special purpose hardware-based systems that perform

the specified functions or acts or carry out combinations of special purpose hardware and computer instructions.

(43) The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the invention. As used herein, the singular forms “a”, “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “comprises” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

(44) The corresponding structures, materials, acts, and equivalents of all means or step plus function elements, if any, in the claims below are intended to include any structure, material, or act for performing the function in combination with other claimed elements as specifically claimed. The description of the present invention has been presented for purposes of illustration and description, but is not intended to be exhaustive or limited to the invention in the form disclosed. Many modifications and variations will be apparent to those of ordinary skill in the art without departing from the scope and spirit of the invention. The embodiment was chosen and described in order to best explain the principles of the invention and the practical application, and to enable others of ordinary skill in the art to understand the invention for various embodiments with various modifications as are suited to the particular use contemplated.

Claims

1. A semiconductor device for a wireless power receiver, the semiconductor device comprising: a power rectifier configured to rectify alternating current (AC) power into direct current (DC) power, the power rectifier including a first high side transistor, a second high side transistor, a first low side transistor, and a second low side transistor; a controller; and a plurality of comparators configured to: measure voltage levels across the first high side transistor, the second high side transistor, the first low side transistor, and the second low side transistor; and output a plurality of signals indicating the measured voltage levels to the controller, wherein the controller configured to: operate the power rectifier in full bridge rectifier mode; detect a misfire event between the wireless power receiver and a wireless power transmitter based on the plurality of signals; and in response to detection of the misfire event, switch an operation mode of the power rectifier from the full bridge rectifier mode to a diode mode.
2. The semiconductor device of claim 1, wherein the controller is configured to switch off the first high side transistor, the second high side transistor, the first low side transistor, and the second low side transistor to operate the power rectifier under the diode mode.
3. The semiconductor device of claim 1, wherein the controller is configured to detect an initial misfire cycle of the misfire event.
4. The semiconductor device of claim 1, wherein under the full bridge rectifier mode of the power rectifier, the controller is configured to: in response to an expiration of a minimum ON time, determine whether a drive signal being used for switching on a current path is HIGH or LOW; in response to determination of the drive signal being HIGH, determine that the misfire event has not occurred; and in response to determination of the drive signal being LOW, identify a potential misfire event.
5. The semiconductor device of claim 4, wherein in response to identification of the potential misfire event, the controller is configured to: determine whether a voltage level measured across a low side transistor of the current path indicates HIGH or LOW; in response to determination of the voltage level indicating HIGH, determine that the potential misfire event has not occurred; and in response to determination of the voltage level indicating LOW, determine that the potential misfire event has occurred.

6. The semiconductor device of claim 4, wherein detection of whether the drive signal is HIGH or LOW is performed in a predetermined time window that begins in response to the expiration of the minimum ON time.
7. The semiconductor device of claim 1, wherein the controller is configured operate the power rectifier in the diode mode for one or more cycles.
8. An apparatus comprising: a plurality of comparators configured to: measure voltage levels across a first high side transistor, a second high side transistor, a first low side transistor, and a second low side transistor of a power rectifier; and output a plurality of signals indicating the measured voltage levels; and a controller configured to: operate the power rectifier in full bridge rectifier mode; based on the plurality of signals outputted by the plurality of comparators, detect a misfire event between the power rectifier and a wireless power transmitter; and in response to the detection of the misfire event, switch an operation mode of the power rectifier from the full bridge rectifier mode to a diode mode.
9. The apparatus of claim 8, wherein the controller is configured to switch off the first high side transistor, the second high side transistor, the first low side transistor, and the second low side transistor to operate the power rectifier under the diode mode.
10. The apparatus of claim 8, wherein the controller is configured to detect an initial misfire cycle of the misfire event.
11. The apparatus of claim 8, wherein under the full bridge rectifier mode of the power rectifier, the controller is configured to: in response to an expiration of a minimum ON time, determine whether a drive signal being used for switching on a current path is HIGH or LOW; in response to determination of the drive signal being HIGH, determine that the misfire event has not occurred; and in response to determination of the drive signal being LOW, identify a potential misfire event.
12. The apparatus of claim 11, wherein in response to identification of the potential misfire event, the controller is configured to: determine whether a voltage level measured across a low side transistor of the current path indicates HIGH or LOW, wherein the voltage level is represented by a signal among the plurality of signals; in response to determination of the voltage level indicating HIGH, determine that the potential misfire event has not occurred; and in response to determination of the voltage level indicating LOW, determine that the potential misfire event has occurred.
13. The apparatus of claim 12, wherein the detection of whether the drive signal is HIGH or LOW is performed in a predetermined time window that begins in response to the expiration of the minimum ON time.
14. A method for operating a wireless power receiver, the method comprising: operating, by a controller, a power rectifier of a wireless power receiver in full bridge rectifier mode; receiving, by the controller, a plurality of signals from a plurality of comparators, wherein the plurality of signals is based on measurements of voltage levels across a first high side transistor, a second high side transistor, a first low side transistor, and a second low side transistor of a power rectifier; based on the plurality of signals, detecting, by the controller, a misfire event between the wireless power receiver and a wireless power transmitter; and in response to the detection of the misfire event, switching, by the controller, an operation mode of the power rectifier from the full bridge rectifier mode to a diode mode.
15. The method of claim 14, wherein switching the operation mode of the power rectifier from the full bridge rectifier mode to the diode mode comprises switching off the first high side transistor, the second high side transistor, the first low side transistor, and the second low side transistor.
16. The method of claim 14, wherein detecting the misfire event comprises detecting, by the controller, an initial misfire cycle of the misfire event.
17. The method of claim 14, wherein under the full bridge rectifier mode of the power rectifier, the method further comprising: in response to an expiration of a minimum ON time, determining, by the controller, whether a drive signal being used for switching on a current path is HIGH or LOW; in response to determining the drive signal being HIGH, determining, by the controller, that the

misfire event has not occurred; and in response to determining the drive signal being LOW, identifying, by the controller, a potential misfire event.

18. The method of claim 17, wherein in response to identifying the potential misfire event, the method further comprising: determining, by the controller, whether a voltage level measured across a low side transistor of the current path indicates HIGH or LOW; in response to determining the voltage level indicating HIGH, determining, by the controller, that the potential misfire event has not occurred; and in response to determining the voltage level indicating LOW, determining, by the controller, that the potential misfire event has occurred.

19. The method of claim 17, wherein detecting whether the drive signal is HIGH or LOW is performed in a predetermined time window that begins in response to the expiration of the minimum ON time.
